

The Effects of Aging Populations on Economic Growth in Developed Countries

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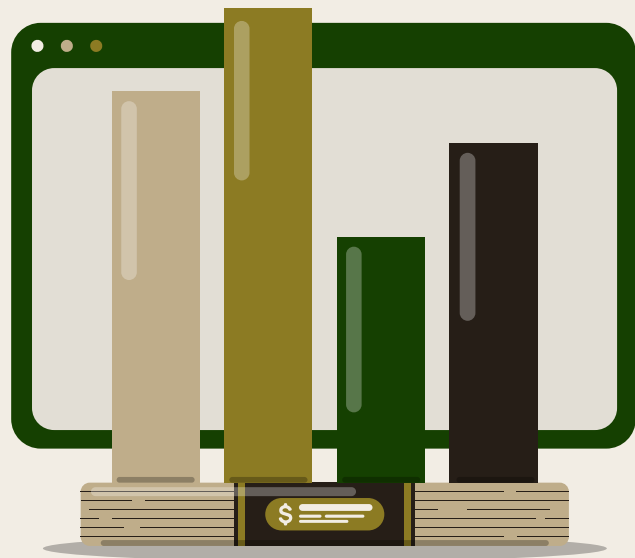
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01

Background on Population Aging in Developed Countries





What is Population Aging?

According to the United Nations, population aging refers to the increasing median age in a country's population due to declining birth rates and rising life expectancy.

Causes of Population Aging



Declining Birth Rates

Many developed countries have fertility rates below the replacement level (2.1 children per woman), leading to a shrinking younger population.



Increased Life Expectancy

Advances in healthcare, nutrition, and living conditions have extended life spans, leading to a larger elderly population and shifting demographic structures worldwide.



Lower Immigration Rates

Some countries rely on immigration to sustain their workforce, but in many cases, migration rates are insufficient to offset population aging and labor shortages.

Global Trends in Aging

Europe & Japan: Countries like Japan, Germany, and Italy have some of the highest proportions of elderly citizens. Japan, for example, has over 28% of its population aged 65 and older.



China & Other Emerging Economies: Some developing nations, like China, are aging rapidly due to past policies (e.g., China's one-child policy), which could create economic challenges.

United States & Canada: Aging is slower in North America but still significant, with the baby boomer generation (born 1946–1964) entering retirement.

Economic And Social Implication



Labor Market Shifts

Fewer young workers mean potential labor shortages and increased demand for automation and older workforce participation.



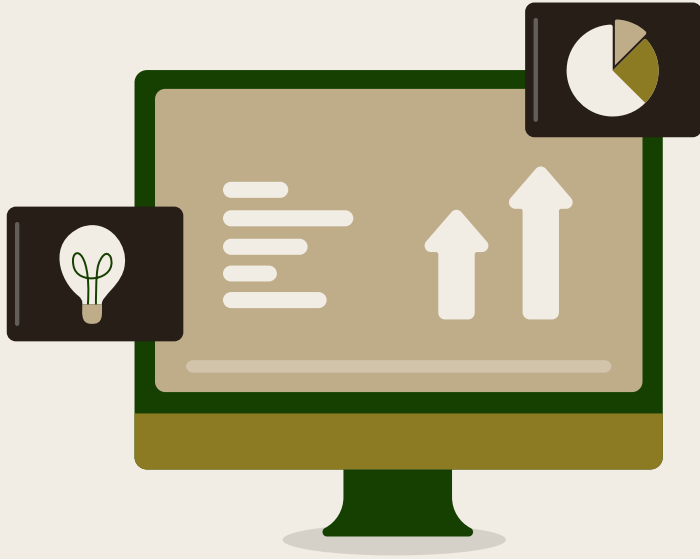
Pension And Healthcare Costs

Aging populations increase public spending on retirement benefits and healthcare, creating fiscal strain.



Intergenerational Challenges

Younger workers may face higher taxes to support an aging population, while economic growth may slow due to lower productivity.



02

Literature Review

Literature Review

The aging population has significant implications for economic growth, primarily through its effects on government expenditure, consumption patterns, and income sources.



Some studies suggest a negative relationship, as an increasing proportion of elderly and young dependents may reduce labor force participation, savings, and capital demand, thereby slowing economic growth



In developed nations, institutional flexibility and policy adaptations help mitigate these adverse effects



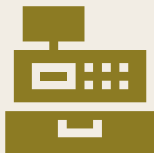
In developing countries, aging populations have not yet significantly hindered economic development due to their relatively smaller elderly demographic and larger working-age populations.



Literature Review

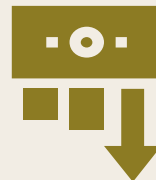
Fiscal Pressures and Public Expenditure

Aging populations strain national finances by increasing public spending pensions and healthcare



Consumption Patterns and Economic Behavior

Population aging also influences consumption trends. In low and middle-income countries, elderly consumption levels decline relative to younger cohorts (aged 30–49).



Labor Market Adjustments

To counter labor force contractions, policies such as raising retirement ages and promoting migration have been proposed.



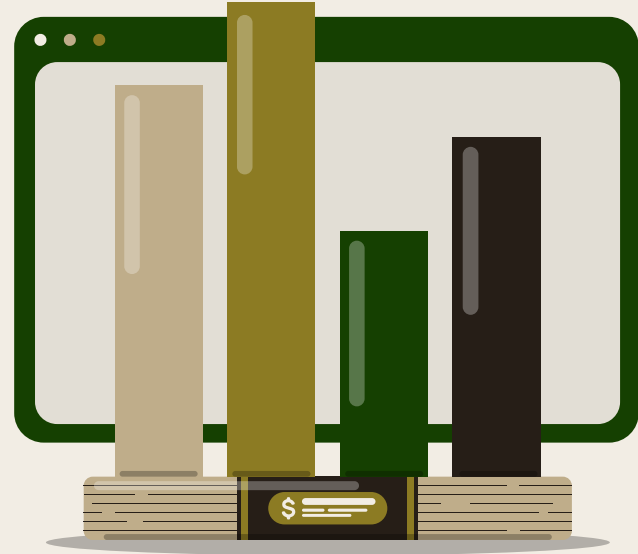
Poverty and Income

In Latin America and the Caribbean, 19% of individuals over 60 live below the poverty line (\$2.50/day)



03

Analysis: The Effects of Aging Populations on Economic Growth in Developed Countries



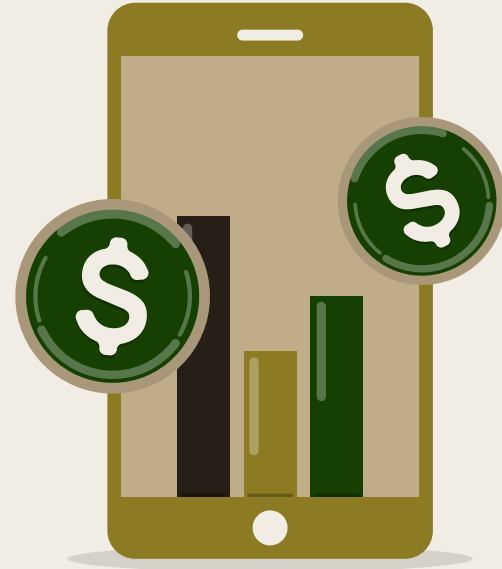
Labor Market Challenges and Workforce Participation

One of the most direct effects of population aging is the shrinking workforce.



Case Study:

Canada has implemented various policies to address workforce challenges associated with an aging population. These include increasing immigration targets to sustain labor force growth, encouraging older workers to remain employed longer, and promoting workforce participation among underrepresented groups such as women and Indigenous peoples.



Labor Market Challenges and Workforce Participation

Current Trends			
Year	Total population (millions)	Working-Age 15-64 (millions)	Percentage of Total Population (%)
2021	38	25.1	66.1%
2025	39.9	26.1	65.4%
2030	41.7	26.3	63%
2035	43.5	26.4	60.7%
2040	45	26.5	58.9%

Fiscal Pressures on Pension and Healthcare Systems

Aging populations place significant strain on public finances, particularly in pension and healthcare spending.

Case Study

In Canada, healthcare spending has surged due to an aging population. According to the Canadian Institute for Health Information (2022), seniors account for nearly 48% of total healthcare spending, despite representing less than 20% of the population.

Case Study		
Age Group	Percentage of population (2022)	Percentage of Healthcare Spending (2022)
0-14	16%	8%
15-64	65%	44%
65+	19%	48%

Fiscal Pressures on Pension and Healthcare Systems

Data Insight		
Year	Public Pension Expenditures (% of GDP)	Healthcare Expenditures Trend
2020	8%	Increasing demand for elder care and chronic disease management
2050	12-16%	Expected surge due to aging population

Changing Consumption Patterns and Economic Behavior

As populations age, consumer spending patterns shift. Older individuals tend to allocate a higher percentage of their income toward healthcare, housing, and services while reducing expenditures on education, transportation, and durable goods.

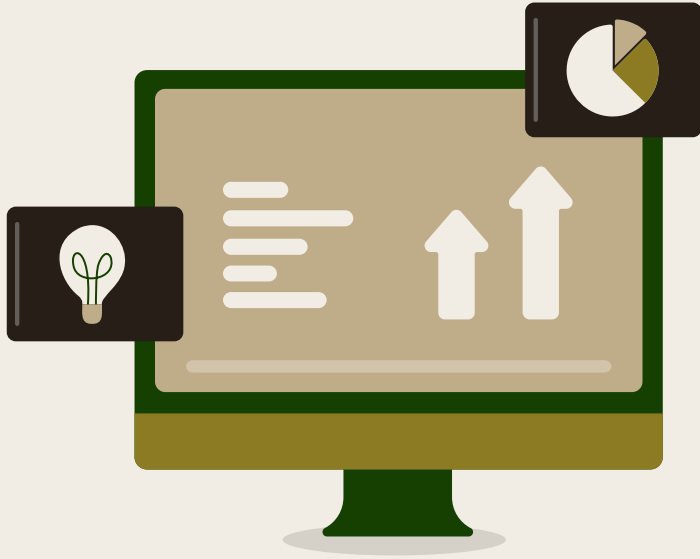


Trends in Consumption: A study by McKinsey & Company found that individuals over 60 control more than 50% of total consumer spending in developed countries.



Case Study

Housing Market Impact: Aging Canadians are increasingly downsizing from larger homes to condominiums or assisted-living facilities, influencing housing demand. This shift is altering real estate market trends, with a growing emphasis on retirement communities.



04

Discussion

Changing Consumption Patterns and Economic Behavior

Variable	Obs.	Mean	Std. dev.	Min	Max
GDP annual growth rate	2229	1.49	5.51	-57.53	25.57
Log GDP per capita	2229	77.91	7.32	57.38	94.05
Capital stock	2229	3.75	14.72	0.01	218.98
Trade Openness	2229	66.89	32.09	0.17	277.14
Life Expectancy	2229	57.57	9.22	28.11	75.70
Education	2229	87.84	26.72	12.15	152.25
Youth-age-share	2229	41.63	6.75	14.10	50.22
Old-age-share	2229	3.83	1.93	1.68	16.05
Δ Capital	2229	0.21	1.23	-0.90	36.89
Δ Trade Openness	2229	0.74	11.79	-99.07	206.83
Δ Life Expectancy	2229	0.42	0.72	-6.23	12.28
Δ Education	2229	1.08	4.96	-34.51	48.51
Δ Youth-age-share	2229	-0.18	0.31	-1.39	0.72
Δ Old-age-share	2229	0.02	0.07	-0.28	0.61

The findings of this study provide critical insights into the relationship between demographic shifts and economic growth, with significant implications for policymakers.

Changing Consumption Patterns and Economic Behavior

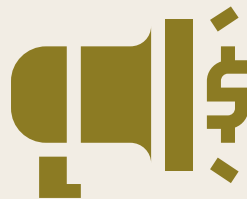
		Old age share	Relative to Average	Youth age share	Relative to Average
Percentile	1	2.17	0.57	17.72	0.43
	5	2.47	0.65	27.19	0.65
	10	2.61	0.68	31.84	0.76
	20	2.79	0.73	37.71	0.90
	30	2.96	0.78	41.07	0.99
	40	3.08	0.81	42.67	1.02
	50	3.24	0.85	43.84	1.05
	60	3.49	0.91	44.71	1.07
	70	3.79	0.99	45.54	1.09
	80	4.32	1.13	46.38	1.11
	90	5.22	1.37	47.60	1.14
	95	6.70	1.75	48.67	1.17
	99	14.21	3.72	49.81	1.20
Average		3.82		41.68	

This table further reveals that the distribution of these age groups varies significantly across percentiles, with the elderly share exhibiting greater dispersion at higher percentiles.

Policy Recommendations

Labor Market Adjustments for Youth Dependency

Governments should implement policies that encourage female labor force participation, such as subsidized childcare and flexible work arrangements, to offset the economic drag from high youth dependency ratios.



Harnessing the Economical Potential of Aging Population

Extending retirement ages and promoting lifelong learning can help retain older workers in the labor force, mitigating labor shortages.



Strengthening Education and Trade Openness

Since both factors consistently show positive growth effects, policymakers should prioritize education reforms and trade liberalization to enhance long-term economic resilience.

Policy Recommendations

Slow-Growth

Slow-growth should focus on mitigating aging-related challenges through healthcare investments and productivity-enhancing technologies.



High-Growth

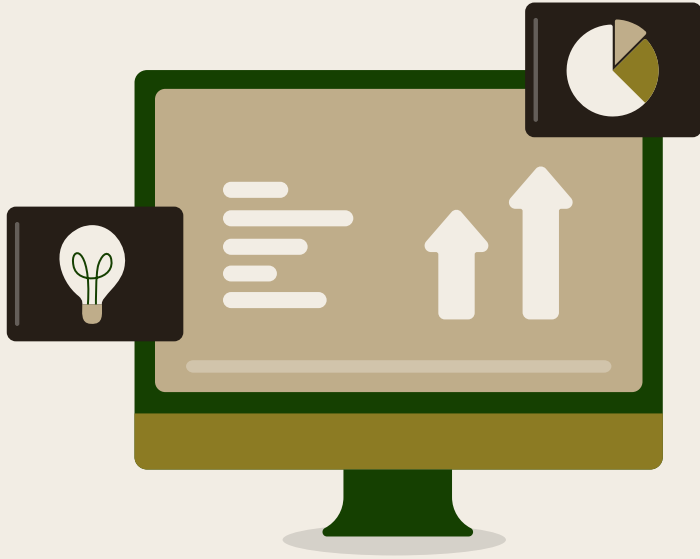
High-growth economies can capitalize on elderly consumption and savings by fostering financial instruments that channel retirement funds into productive investments.



Future Research Directions

Further studies should explore regional variations in demographic impacts, particularly how urbanization and technological adoption mediate these effects. Additionally, the role of automation in compensating for labor shortages due to aging populations warrants deeper investigation.





05

Conclusion

CONCLUSION

Canada's aging population is reshaping key aspects of the economy, labor force, and social infrastructure. By 2050, public pension expenditures are projected to rise to 12–16% of GDP, and healthcare costs will continue surging due to increased demand for elder care.



The labor market is also transforming. Canada has responded through policies such as raising immigration targets and promoting workforce participation among underrepresented groups to counteract labor shortages.



From a consumer perspective, older Canadians control a significant share of total spending, yet they have lower spending elasticity, potentially reducing overall economic dynamism. This has implications for market trends, as businesses may need to adapt their products and services to meet the needs of an aging consumer base.



In the housing sector, seniors are increasingly downsizing from large homes to condominiums or assisted-living facilities, altering real estate trends. This shift is driving growth in retirement communities and senior housing investments, while also reducing demand for single-family homes.

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THANKS!

Do you have any questions?